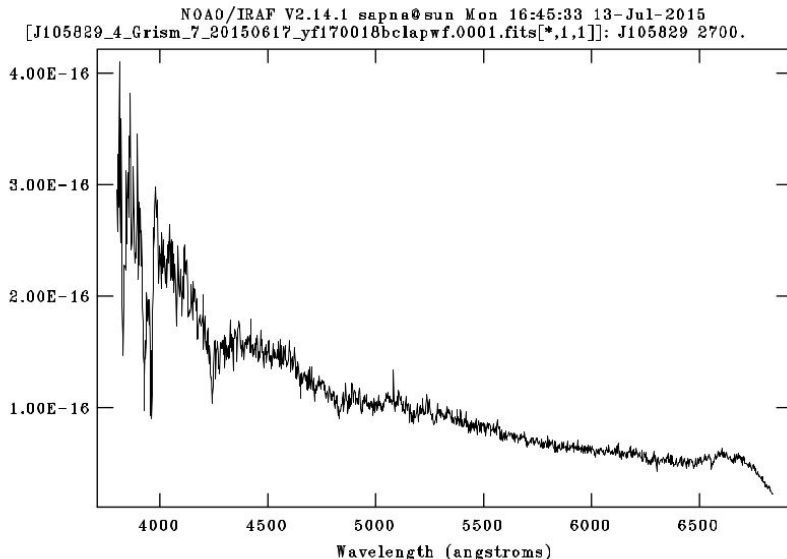


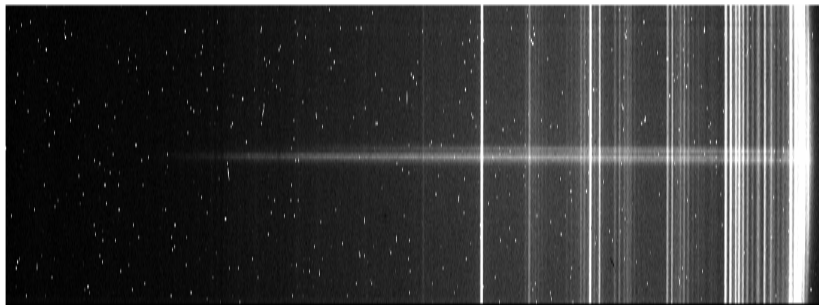
Spectroscopy: concepts and data analysis

Sapna Mishra
Aryabhata Research Institute of Observational Sciences

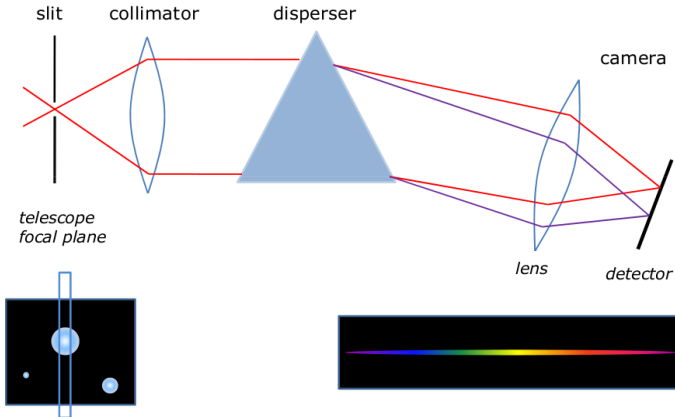
March 20, 2018

Expectation...





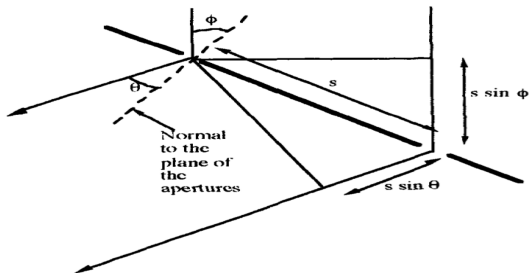
The basic spectrograph



Components of spectroscopy

- Slits
 - To increase resolution
 - To block lights from unwanted portion of Sky
- Collimator
 - To make diverging light from the slit parallel and directs the collimated beam towards the grating
 - $n_{fixed} \lambda_{fixed} = d(\sin \theta_i + \sin \theta_d)$
 - Otherwise same wavelength in same order would be imaged onto different positions on the detector, blurring the resulting spectrum.
- Disperser
 - To disperse light into its wavelength components
 - prisms, gratings
- Camera
 - To collect the spectrally-dispersed collimated beams from the grating, and make them converge onto the detector.
- Detector
 - To image the spectrum

Grating Equation and Dispersion



$$n\lambda = s(\sin \phi + \sin \theta)$$

where n : order of maximum intensity

s : distance between two slits

ϕ : angle of incidence normal to grating

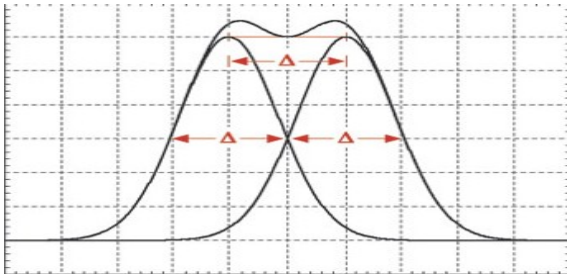
θ : angle of diffraction

Dispersion: Splitting of light into its components, or rate of change of wavelength with angular position.

$$\frac{d\lambda}{d\theta} = \frac{s \cos \theta}{n}$$

Spectral Resolution

The spectral resolution or spectral resolving power, R , of a spectrograph is defined as the ability to distinguish between two wavelengths separated by a small amount $\Delta\lambda$.

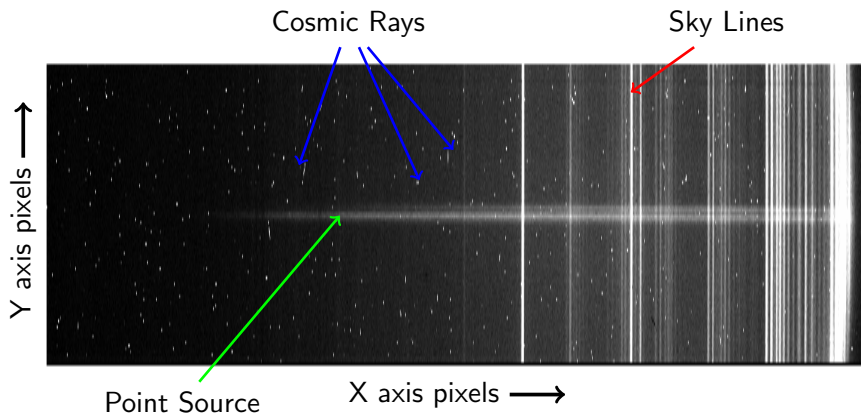


$$R = \frac{\lambda}{\Delta\lambda} = nN$$

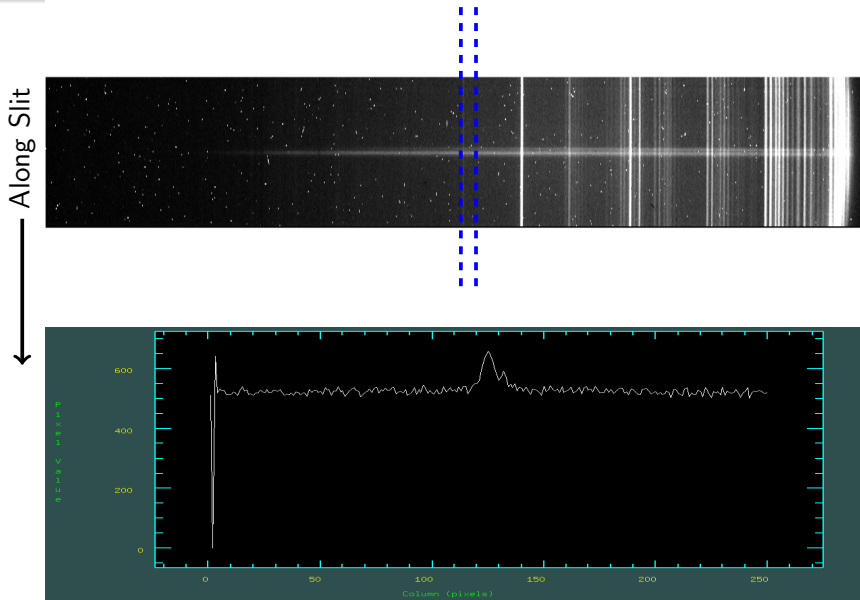
where n : order of maximum intensity

N : number of aperture

What are they????

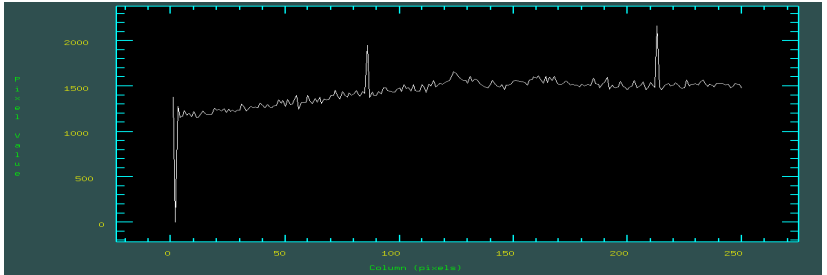
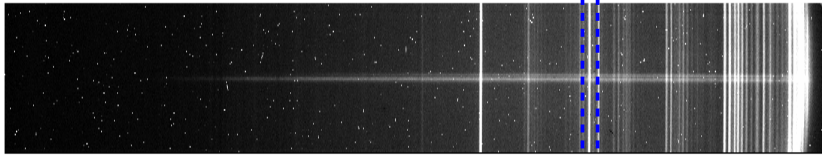


The detector view ; spatial slice

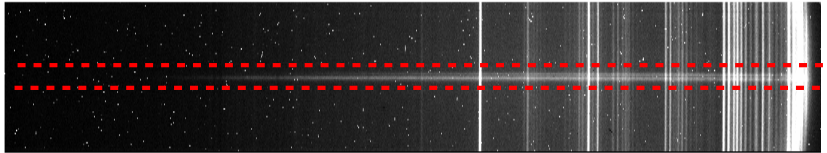


The detector view ; spatial slice

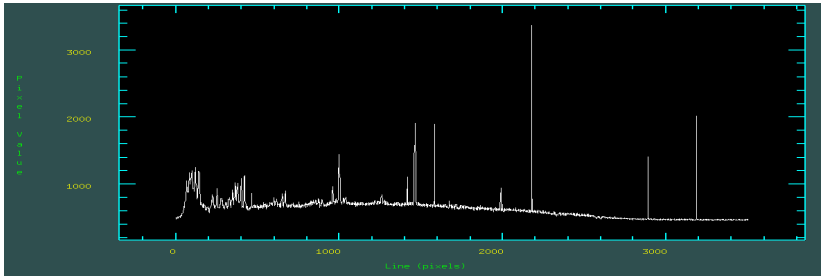
Along Slit



The detector view ; spectral slice



dispersion direction →



Slit Optimization



Resolution degrades by slit width



Seeing optimized slit width



Light loss due to seeing

Steps involved in spectroscopic reduction

- Bias Subtraction
- Flat correction
- Cosmic ray removal
- Aperture tracing and spectral extraction
- Wavelength Calibration
- Flux Calibration

What are bias, why and how to correct??

- Unwanted counts arising from limitations of electronics.
- Bias are zero exposure frames of CCD
- Combine all bias frames using median

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- Unwanted counts arising from limitations of electronics.
- Bias are zero exposure frames of CCD
- Combine all bias frames using median ???????
- zerocombine task under IRAF is used to combine them.
- Pixel to pixel median is taken to create master bias (in IRAF!!!)

```
ccdred>
ccdred>
ccdred> !ls Bias_250_3500* > bias.in
ccdred> imstat @bias.in
```

#	IMAGE	NPIX	MEAN	STDEV	MIN	MAX	
	Bias_250_3500_8_Free_yf170021.fits	875000	459.7	33.52	0.	7440.	
	Bias_250_3500_8_Free_yf170022.fits	875000	459.7	33.13	0.	9106.	
	Bias_250_3500_8_Free_yf170023.fits	875000	459.9	33.48	0.	6876.	
	Bias_250_3500_8_Free_yf170035.fits	875000	476.1	32.34	0.	3027.	
	Bias_250_3500_8_Free_yf170036.fits	875000	476.1	33.99	0.	6351.	
	Bias_250_3500_8_Free_yf170037.fits	875000	476.	32.15	0.	3095.	

```
ccdred> █
```


Zerocombine

```
PACKAGE = ccdred
TASK = zerocombine

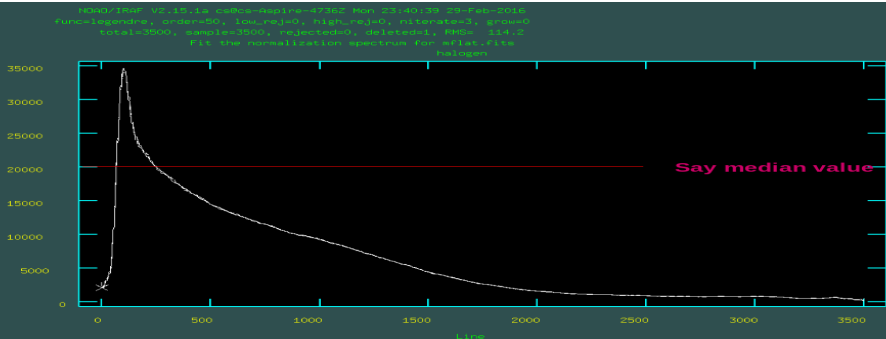
input = @bias.in List of zero level images to combine
(output = mbias.fits) Output zero level name
(combine= median) Type of combine operation
(reject = minmax) Type of rejection
(ccdtype= ) CCD image type to combine
(process= no) Process images before combining?
(delete = no) Delete input images after combining?
(clobber= no) Clobber existing output image?
(scale = none) Image scaling
(statsec= ) Image section for computing statistics
(nlow = 0) minmax: Number of low pixels to reject
(nhigh = 1) minmax: Number of high pixels to reject
(nkeep = 1) Minimum to keep (pos) or maximum to reject
(mclip = yes) Use median in sigma clipping algorithms?
(lsigma = 3.) Lower sigma clipping factor
(hsigma = 3.) Upper sigma clipping factor
(rdnoise= 4.8) ccdclip: CCD readout noise (electrons)
(gain = 1.22) ccdclip: CCD gain (electrons/DN)
(snoise = 0.) ccdclip: Sensitivity noise (fraction)
(pclip = -0.5) pclip: Percentile clipping parameter
(blank = 0.) Value if there are no pixels
(mode = ql)
```

Flat Correction

- Unwanted counts arising from pixel to pixel sensitivity variation of detector.
- Flats are uniformly illuminated fields , (dome flats)
- Again combine all flat frames using median (same reason)
- flatcombine task under IRAF is used to combine them.
- Wavelength to wavelength sensitivity should be corrected.

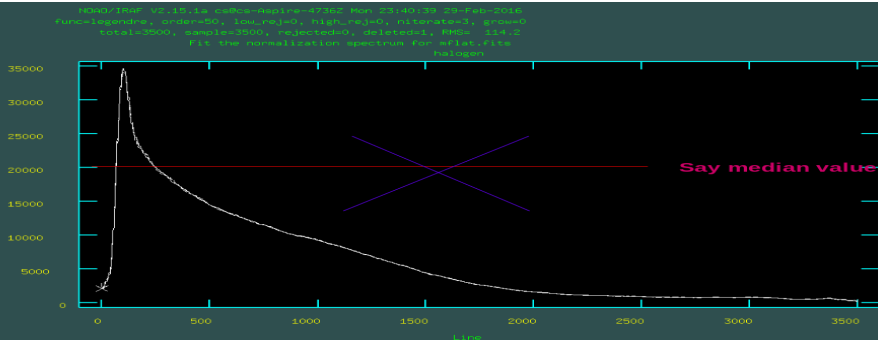
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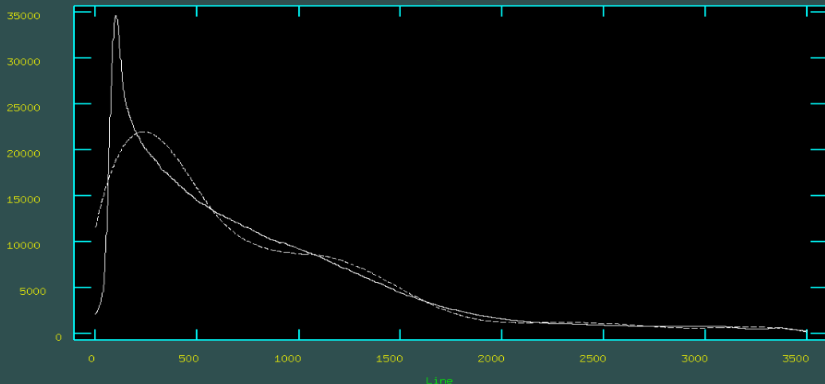


Flat Correction

Dotted Curve: Instrumental Curve (To fit)

Solid Curve: Observed Curve (mflat.fits)

```
NOAO/IRAF V2.15.1a cs@cs-Repire-47362 Thu 17:26:17 03-Mar-2016
func=spline3, order=6, low_rej=0, high_rej=0, niterate=1, grow=0
total=3500, sample=3500, rejected=0, deleted=0, RMS= 2288.
Fit the normalization spectrum for mflat.fits
halogen
```



Output will result in normalized flat spectrum (nmflat.fits)

Cosmic Ray removal

- Unwanted counts arising from high energy particles.
- Give rise to sudden increase in counts and hence can be identified as single pixel dirac delta peak in spectrum.
- Task called LACOS SPEC is used to remove them.

```
noao> stsdas
      analysis.      describe      fitsio.      hst_calib.      problems      toolbox.
      contrib.      examples      graphics.      playpen.      sobsolete.
```

```
stsdas> task lacos_spec=/scisoft/share/iraf/extern/stsdas/lacos_spec.cl
stsdas> epar lacos
```

```
stsdas> epar lacos
```

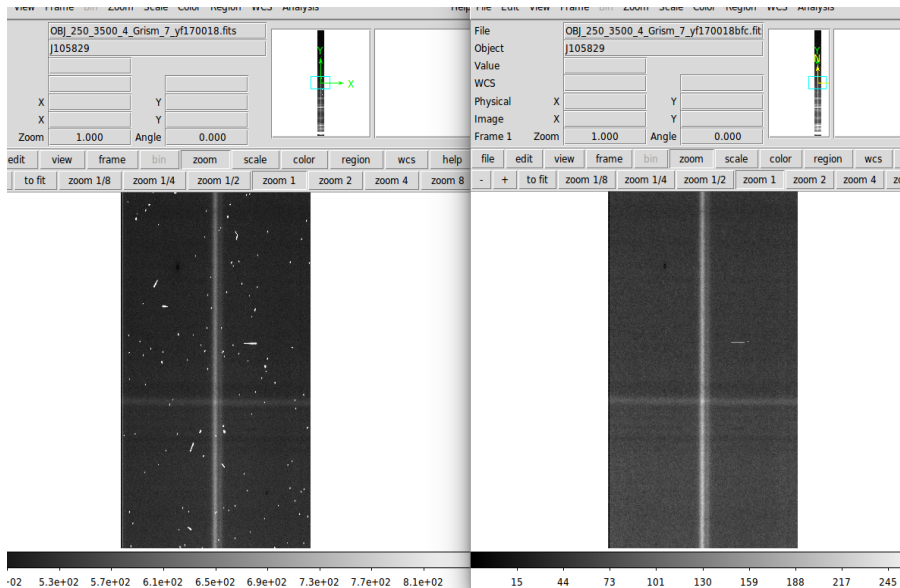
IRAF

Image Reduction and Analysis Facility

```
PACKAGE = stsdas
TASK = lacos_spec
```

```
input   = OBJ_250_3500_4_Grism_7_yf170018bf.fits  input spectrum
output  = OBJ_250_3500_4_Grism_7_yf170018bfc.fits  cosmic ray cleaned output spectrum
outmask = OBJ_250_3500_4_Grism_7_yf170018bf      output bad pixel map (.pl)
(gain   = 1.22) gain (electrons/ADU)
(readn  = 4.8) read noise (electrons)
(xorder = 9) order of object fit (0=no fit)
(yorder = 4) order of sky line fit (0=no fit)
(sigclip= 2.) detection limit for cosmic rays (sigma)
(sigfrac= 0.5) fractional detection limit for neighbouring pixels
(objlim = 1.) contrast limit between CR and underlying object
(niter  = 10) maximum number of iterations
(verbose= yes)
(mode   = al)
```

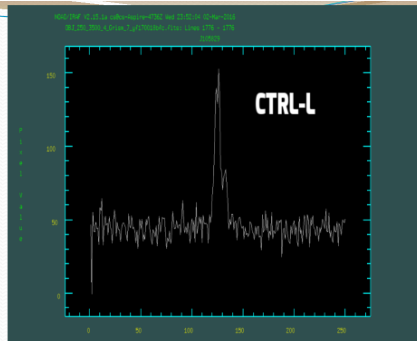
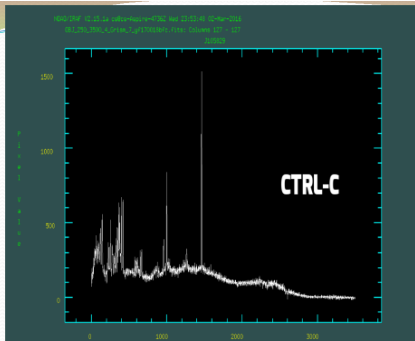
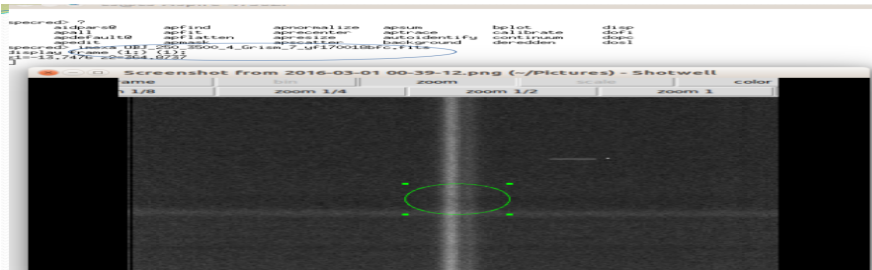
Cosmic Ray removal



Aperture tracing and spectrum extraction

- Identify the source
- Define the aperture
- Subtract the background
- Trace the aperture
- Apall task under IRAF performs all the above tasks

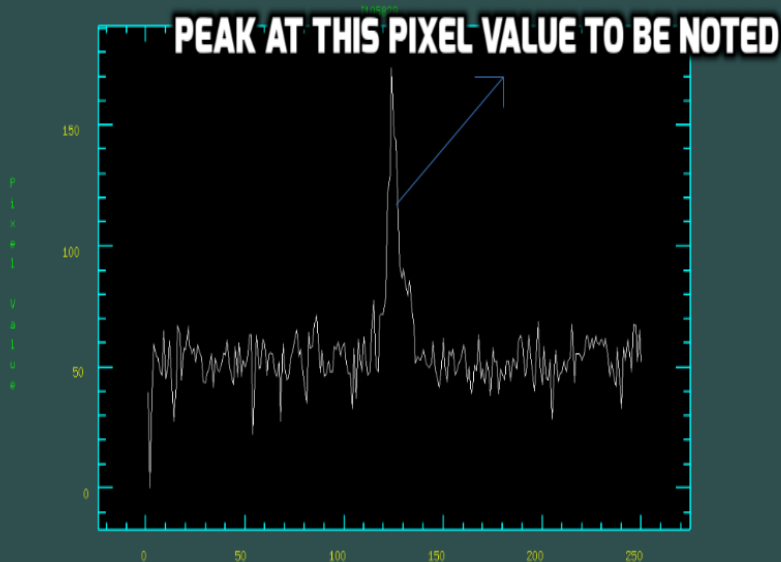
How to identify source



How to identify source

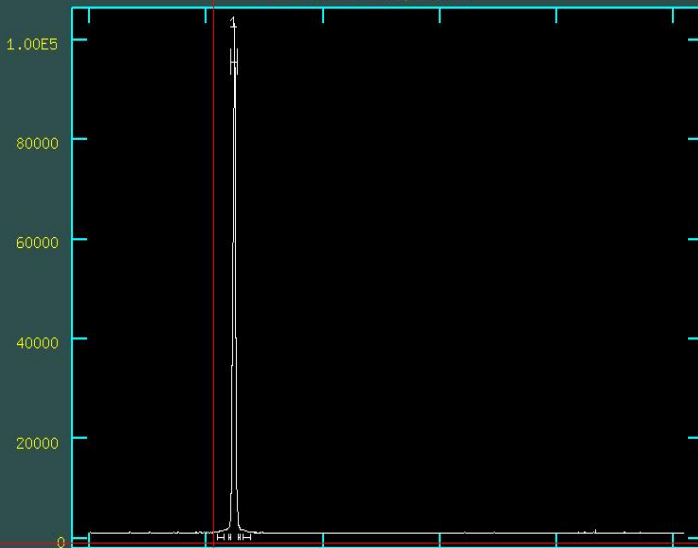
HD40/TRAF V2.15.1a cs@ce-Aspire-4736Z Tue 00:09:55 01-Mar-2016

OBJ_250_3500_4_0rise_7_yf170010bfc.fits: Lines 1729 - 1729



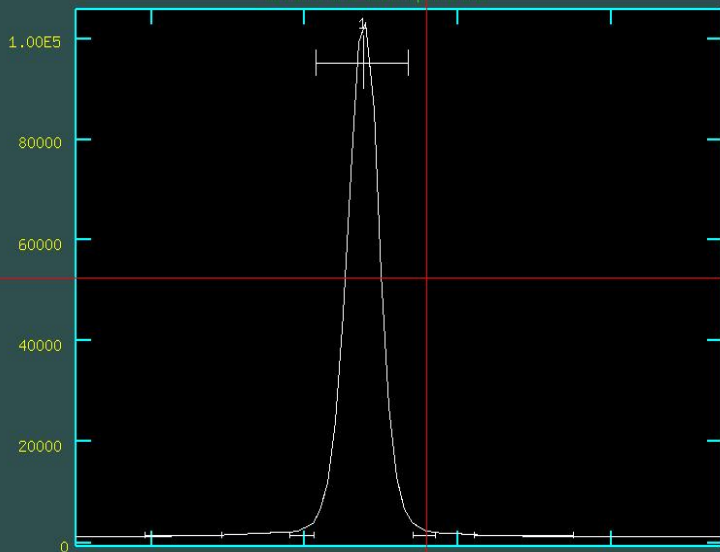
Apall:Defining the Aperture

DAO/IRAF V2.12.2a-EXPORT walawend@trapezium.Colorado.EDU Thu 13:36:55 14-Oct-200
Image=vegal, Sum of columns 760-769
Define and Edit Apertures



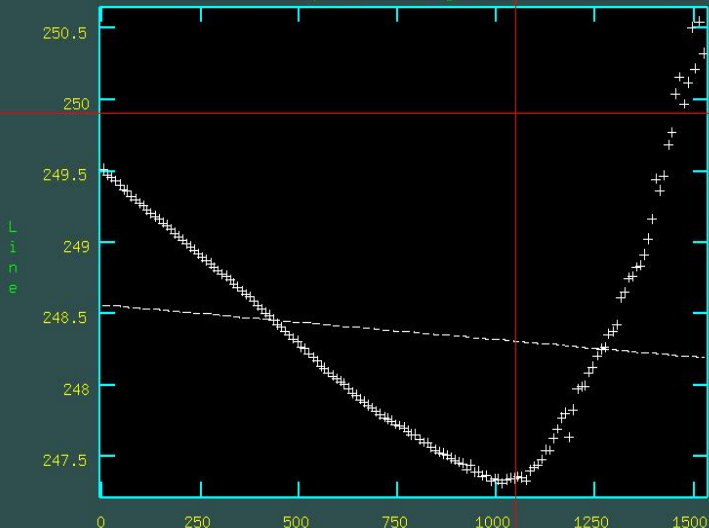
Apall:Subtract Background

AD/IRAF V2.12.2a-EXPORT walawend@trapezium.Colorado.EDU Thu 13:44:38 14-Oct-200
Image=vegal, Sum of columns 760-769
Define and Edit Apertures



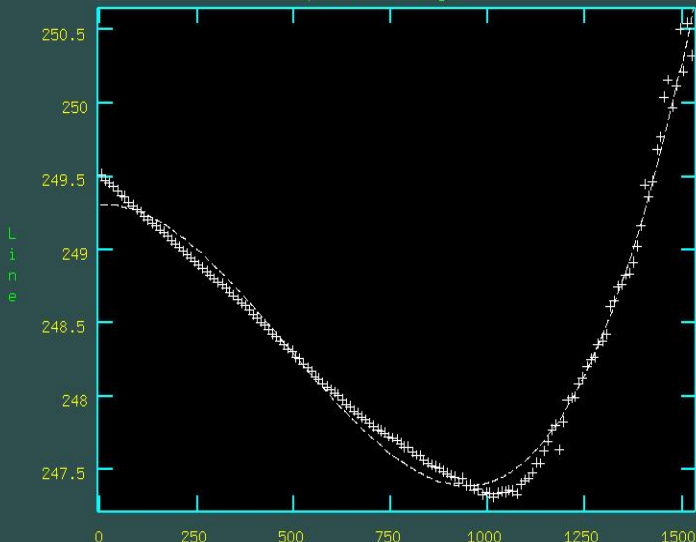
Apall:Defining the Trace

```
AD/IRAF V2.12.2a-EXPORT walawend@trapezium.Colorado.EDU Thu 14:59:03 14-Oct-200  
func=legendre, order=2, low_rej=3, high_rej=3, niterate=0, grow=0  
total=153, sample=153, rejected=0, deleted=0, RMS= 0.7918  
Aperture 1 of vega1
```



Apall:Defining the Trace

```
DAO/IRAF V2.12.2a=EXPORT walawend@trapezium.Colorado.EDU Fri 09:55:30 15-Oct-200  
func=legendre, order=4, low_rej=3, high_rej=3, niterate=0, grow=0  
total=153, sample=153, rejected=0, deleted=0, RMS= 0.0912  
Aperture 1 of vegal
```



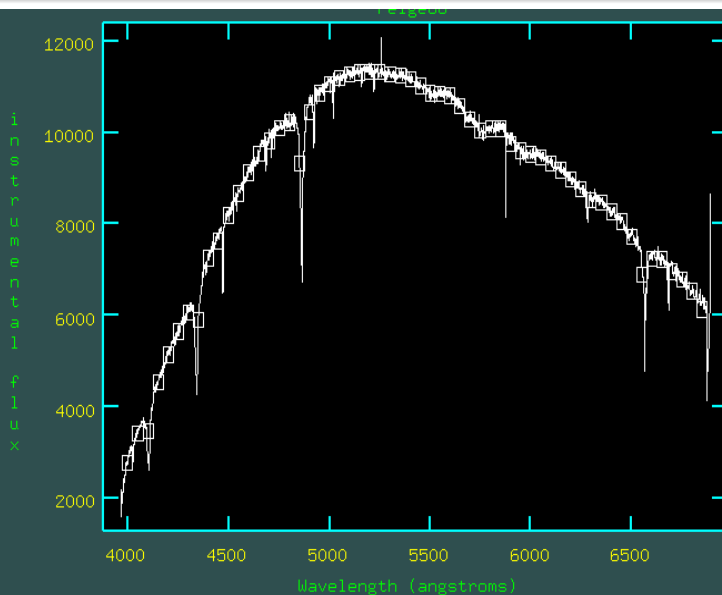
Wavelength Calibration: pixel to wavelength mapping

- IRAF tasks involved: identify, hedit, dispcor
- identify : to get pixel to wavelength relation
- hedit: prepare the solution for science object
- dispcor: apply the solution to science object

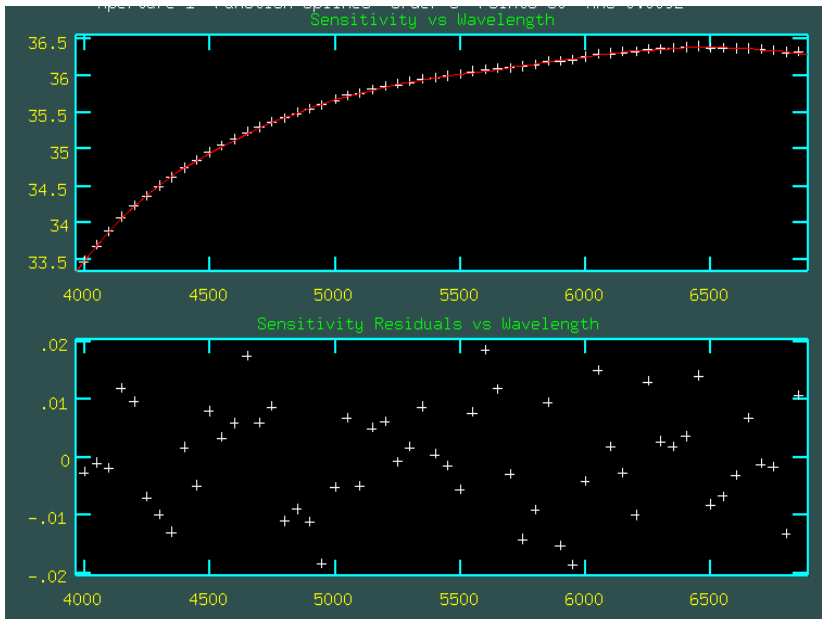
Flux Calibration: Conversion from counts to known units of flux

- IRAF tasks involved: standard, sensfunc, calibrate
- standard: tabulate the observed count and extinction magnitude
- sensfunc: calculate the sensitivity curve
- calibrate: apply the sensitivity curve on to the science object

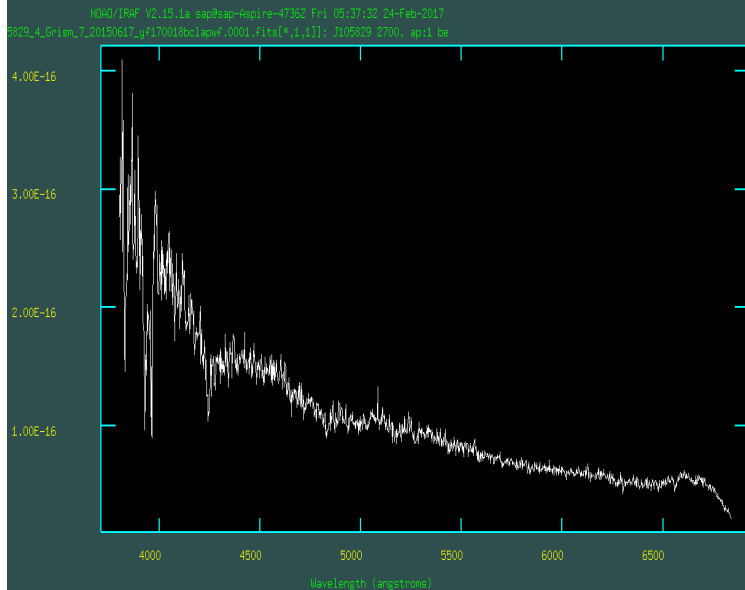
standard plot



Sens plot



Final Flux calibrated spectrum



What next???

- Heliocentric correction (iraf task rvcorrect)
- Air to vaccum correction (iraf task disptrans)
- Calculate the composition, ionization state, temperature, velocity, density, equivalent width, column density of your source.
- Enjoy!!!!

Thank You !!!!